Research Highlights

- Four packet-level attacker actions are modeled as Lie-cone generators on flows.
- Trajectories in the cone provably preserve rate, length, header and MTU invariants.
- FFE reaches 59.0% Real ASR on real CSE-CIC-IDS2018 with 0.392 perturbation cost.
- Physical feasibility reaches 100% while mathematical baselines collapse to 0–1%.
- Dual defense (MPS + LGAT) slashes FFE success from 51.6% to 5.3% at M=5.